



VADODARA INSTITUTE OF ENGINEERING

Lesson Planning – Theory

Subject: Object Oriented Programming

Discipline: CE

Academic Year: 2025-2026

Employee code: 25CH0587

Subject Code: DI04000021

Semester: 4th

Name of Faculty: Prof. Naman Shah

Total no of lecture planned: 42

Teaching and Examination Scheme:

Teaching Scheme			Credits	Examination Marks					Total Marks
L	T	P	C	Theory Marks		Practical Marks			
				ESE (E)	PA (M)	ESE (V)		PA (I)	
					PA	ESE	OEP		
3	0	2	4	70	30	20	10	20	150

L- Lectures, T-Tutorial, P-Practical; C-Credit; ESE- End Semester Exam; PA- Progressive Assessment

Lecture	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Turn on:	√			√		√

Lect. No.	Topic Covered	Book No.	Page No.	CT1	CT2	MSE/ RMSE	Planned Date	Actual Date	HOD's Sign	Remarks
1	Introduction to Java: history, features, and applications	1	1	√		√	03/01			
2	JDK, JRE, JVM: differences and roles; Bytecode explained	1	6	√		√	05/01			
3	Structure of a Java program: class, main() method	1	7	√		√	06/01			
4	Hello World program: writing, compiling, and running	1	9	√		√	07/01			
5	Primitive Data Types: byte, short, int, long, float, double, char, boolean	1	15	√		√	10/01			
6	Declaring & initializing variables; Scope and Lifetime of Variables	1	16	√		√	16/01			
7	Type Conversion and Type Casting	1	18	√		√	17/01			
8	Operators: Arithmetic, Bitwise, Relational, Logical	1	19	√		√	19/01			
9	Operators: Assignment, Conditional, Ternary, Increment and Decrement	1	21	√		√	20/01			
10	Control Flow: if-else (simple, nested, ladder)	1	32,33	√		√	21/01			
11	switch-case statements	1	35,36	√		√	24/01			
12	Loops: for, while, do-while	1	37,38	√		√	27/01			
13	Jump Statements: break and continue	1	59	√		√	28/01			
14	1D Arrays: declaration, initialization, traversal	1	66	√		√	31/01			
15	2D Arrays: introduction and basic operations	1	67		√	√	02/02			
16	Problem Solving Practice: Data Types, Control Flow & Arrays	1	72		√	√	03/02			
17	Principles of OOP: Abstraction and Encapsulation	1	83		√	√	04/02			
18	Principles of OOP: Inheritance and Polymorphism	1	118		√	√	09/02			
19	Classes & Objects: fields and methods	1	119		√	√	10/02			
20	Constructors: Default, Parameterized, and Constructor Overloading	1	140		√	√	16/02			



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21	Important Keywords: this, static (class, block, method, variable)	1	151		√	√	17/02			
22	Access Modifiers: public, private, protected, default	1	158		√	√	18/02			
23	Methods: definition, calling, and Method Overloading	1	154		√	√	21/02			
24	String Class and common String methods	1	175		√	√	23/02			
25	Wrapper Classes: Byte, Short, Integer, Long, Float, Double, Character, Boolean	1	159		√	√	24/02			
26	Autoboxing and Unboxing	1	179		√	√	25/02			
27	User Input: Scanner Class and Command Line Arguments	1	186		√	√	28/02			
28	Garbage Collection and finalize() method	1	180		√	√	02/03			
29	Problem Solving Practice: OOP Concepts	1	213		√	√	03/03			
30	Inheritance: concept, need, and types	1	246		√	√	04/03			
31	super keyword, Method Overriding, and Dynamic Method Dispatch	1	247		√	√	07/03			
32	Final Keyword; Abstract Classes and Abstract Methods	1	251		√	√	09/03			
33	Interfaces: declaration, implementation, and Multiple Inheritance	1	256		√	√	10/03			
34	Default and Static Methods in Interface	1	264		√	√	11/03			
35	Packages: defining, importing, and built-in packages	1	271		√	√	14/03			
36	Problem Solving Practice: Inheritance, Interfaces & Abstraction	1	282		√	√	16/03			
37	Types of Errors; Exception Hierarchy	1	290		√	√	17/03			
38	Exception Handling: try, catch, finally, throw, throws; Multiple catch blocks	1	290		√	√	18/03			
39	Built-in Exceptions; Collections Framework: List (ArrayList), Set (HashSet), for-each loop	1	314		√	√	23/03			
40	Modern Java Features: Lambda Expressions, Stream API; Introduction to AI with Java (DL4J, TensorFlow Java API)	1	314		√	√	24/03			
41	Revision	1	350		√	√	25/03			
42	Revision	1	350		√	√	01/04			
43	Revision	-	-		√	√	04/04			
44	Revision	-	-		√	√	06/04			

Reference Books:

1. Advanced Java, M. T. Savaliya, dreamtech
2. Black Book “Java server programming” J2EE, 1st ed., Dream Tech Publishers, 2008.
3. Kathy walrath
3. Complete Reference J2EE by James Keogh mcgraw publication

Prof. Naman Shah
Subject Coordinator

Prof. Jemisha Patel
Head of Department

Dr. Jayeshkumar S.Patel
Principal